

SEROV, Russian Federation

WMO: 280440

Lat: 59.600N

Lon: 60.533E

Elev: 427

StdP: 14.47

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.8	-21.3	-32.2	0.9	-26.3	-26.5	1.3	-21.1	18.0	-1.3	15.7	4.6	4.8	0	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	84.6	67.0	81.2	65.5	77.7	63.7	69.0	80.8	67.3	77.9	65.6	75.0	4.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	94.4	72.8	63.5	89.2	71.5	61.8	83.7	70.4	33.4	81.6	32.0	77.8	30.7	75.2	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	14.2	12.0	-31.9	89.3	6.1	2.9	-36.3	91.4	-39.9	93.1	-43.3	94.7	-47.8	96.8
			-31.5	70.9	6.2	1.4	-35.9	71.9	-39.5	72.8	-43.0	73.6	-47.5	74.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.9	3.0	7.5	23.9	36.8	50.4	59.8	63.7	59.6	48.4	36.3	19.7	8.3
	DBStd	23.51	13.05	14.22	9.43	8.85	9.36	8.60	7.48	7.47	7.10	9.10	13.02	14.81
	HDD50	6744	1458	1190	808	400	116	18	3	7	112	433	908	1291
	HDD65	11132	1923	1610	1273	845	462	200	117	201	497	890	1358	1756
	CDD50	1248	0	0	0	5	127	312	427	303	66	8	0	0
	CDD65	161	0	0	0	0	8	44	76	32	1	0	0	0
	CDH74	1614	0	0	0	0	112	432	776	289	5	0	0	0
	CDH80	408	0	0	0	0	10	90	220	88	0	0	0	0
Wind	WSAvg	4.8	3.9	4.6	5.6	5.8	5.5	5.2	4.3	4.3	4.6	5.1	4.8	4.2
Precipitation	PrecAvg	19.8	0.9	0.6	0.8	0.9	1.9	3.0	3.2	2.8	2.2	1.2	1.1	0.8
	PrecMax	26.0	1.6	1.2	1.8	2.2	3.5	5.9	8.9	7.3	4.5	2.5	2.5	1.7
	PrecMin	15.2	0.1	0.2	0.0	0.2	0.6	0.5	0.7	0.9	0.4	0.4	0.1	0.2
	PrecStd	2.7	0.4	0.3	0.6	0.5	0.8	1.5	1.6	1.6	1.3	0.6	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	38.1	49.8	65.6	81.3	86.0	88.8	87.8	74.4	62.7	45.8	37.2
		MCWB	30.0	32.6	39.0	51.3	60.1	66.1	69.1	68.9	60.5	52.0	41.6	33.5
	2%	DB	28.0	34.1	43.3	58.1	77.2	82.6	85.4	81.6	68.1	57.3	41.4	34.6
		MCWB	25.8	30.8	35.7	46.5	58.6	65.6	67.7	67.7	57.0	49.5	38.6	31.4
	5%	DB	24.3	30.8	40.0	54.3	72.8	79.4	82.2	76.1	63.2	53.1	37.7	31.1
		MCWB	22.7	27.9	33.3	43.5	56.1	64.2	66.7	64.5	55.0	47.1	34.9	28.6
	10%	DB	20.7	27.1	36.7	50.3	67.9	75.3	78.3	72.1	59.4	49.4	34.8	27.5
		MCWB	19.0	24.2	31.6	41.6	53.6	61.7	64.8	62.5	53.5	44.7	32.1	25.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	33.5	40.1	52.8	62.6	69.9	71.1	70.3	61.9	53.1	42.3	34.0
		MCDB	33.4	36.4	47.1	63.7	75.4	82.2	83.5	84.2	71.5	59.1	44.7	36.4
	2%	WB	26.1	31.0	36.5	47.8	59.9	67.5	69.4	68.0	58.5	50.1	38.9	31.4
		MCDB	27.9	34.1	42.1	55.8	73.6	78.3	81.0	78.9	64.6	56.0	41.2	34.2
	5%	WB	22.7	27.9	34.2	44.5	57.7	65.6	68.0	65.8	56.3	47.3	35.2	28.6
		MCDB	24.3	30.8	38.7	53.1	70.7	76.6	79.0	74.9	61.4	52.9	37.4	31.0
	10%	WB	19.1	24.5	32.1	42.2	54.9	63.7	66.5	63.7	54.1	44.8	32.3	25.5
		MCDB	20.7	27.1	36.2	49.2	65.9	73.4	76.1	70.3	58.6	49.3	34.2	27.5
Mean Daily Temperature Range	5% DB	MDBR	11.7	15.4	15.7	15.7	19.9	17.9	18.0	15.9	13.9	11.2	10.4	11.0
		MCDBR	10.2	13.9	17.5	22.1	26.7	23.5	24.2	23.1	19.8	16.6	9.7	9.4
		MCWBR	9.2	11.1	11.8	13.8	13.1	10.2	8.6	10.2	10.4	10.1	7.9	8.2
	5% WB	MCDBR	10.0	12.6	15.1	20.6	24.9	21.8	21.2	20.7	16.0	15.6	9.5	9.5
		MCWBR	9.2	10.3	11.1	13.6	13.5	10.5	8.5	10.0	9.1	10.1	8.1	8.5
Clear-Sky Solar Irradiance	taub	0.266	0.300	0.354	0.392	0.392	0.392	0.413	0.415	0.367	0.329	0.281	0.253	
	taud	2.029	2.057	1.999	2.102	2.302	2.360	2.302	2.267	2.367	2.334	2.131	1.972	
	Ebn at Noon	156	215	238	253	265	267	257	245	239	210	151	111	
	Edh at Noon	16	26	38	42	37	35	37	35	27	20	14	11	
All-Sky Solar Radiation	RadAvg	142	420	826	1267	1586	1674	1698	1242	735	370	155	75	
	RadStd	14	45	77	134	135	128	144	125	90	50	22	10	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page